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Progress Report

Description of Data

The data I will be using is compiled of 1000 patients and five variables which are typically associated with PCOS diagnosis and its risks. These variables are age, BMI, menstrual irregularity, testosterone level, and number of antral follicles. Finally, the response variable is whether a patient has been diagnosed with PCOS or not, making it a classification data set.

<u>Variable</u>	<u>Description</u>
Age	The age of the patient in years
BMI	$\text{weight(kg)} / \text{height(m)}^2$
Menstrual irregularity	0 means the patient has regular menstruation and 1 means they have irregular menstruation. Irregular menstruation could be classified as missing periods, heavy bleeding, or painful periods.
Testosterone level	Measured as nanograms per deciliter (ng/dL)
Antral follicle count	Antral follicles are the follicles on the patient's ovaries that contain immature eggs. This is the count of how many the patient has at the time of diagnosis.
PCOS diagnosis	0 means they were not diagnosed with PCOS while 1 means they were.

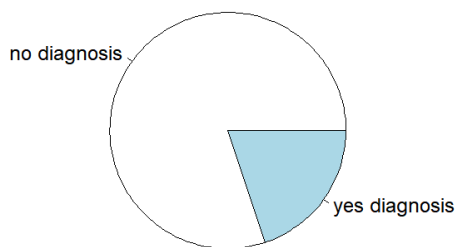
Processing Data

Range of variables:

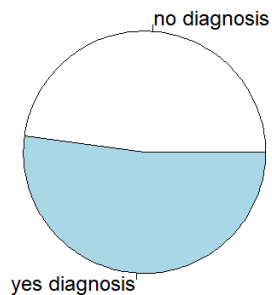
Age	18.0	45.0
BMI	18.2	35.0
Menstrual_Irregularity	0.0	1.0
Testosterone_Level.ng.dL.	20.1	99.6
Antral_Follicle_Count	5.0	29.0
PCOS_Diagnosis	0.0	1.0

Checking for NA's and balancing data:

After running a few tests, I see that I have no missing data throughout the 1000 entries and 6 variables. I do, however, have unbalanced data, 80.1% with no diagnosis and 19.9% with a diagnosis. Having such a large unbalance in data can lead to biased outcomes when trying to model the data.



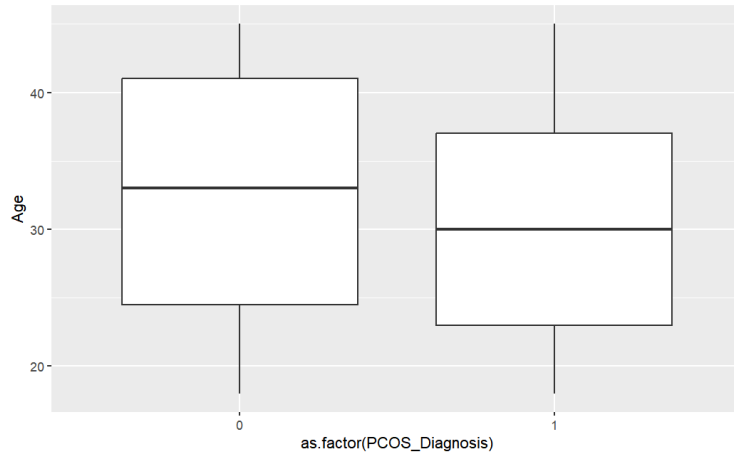
Now I have more balanced data with only 382 entries, 183 (47.9%) with no diagnosis and 199 (52.1%) with a diagnosis. Also, the range of the cut-down data is the same as the original. So, this data will be a great sample that does not lead to bias.



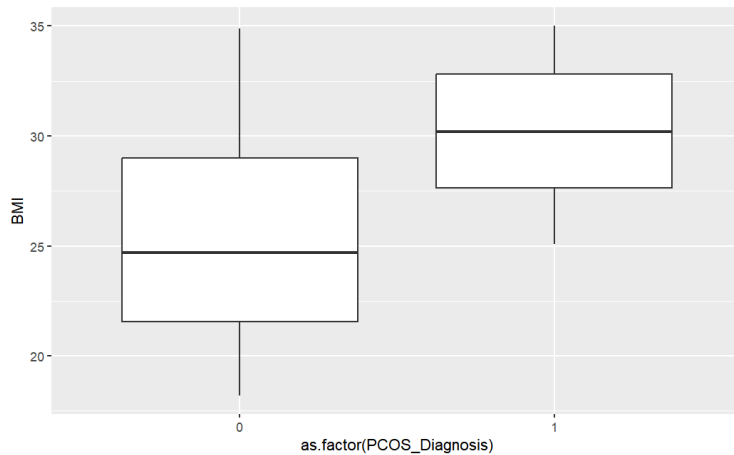
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Data Visualization

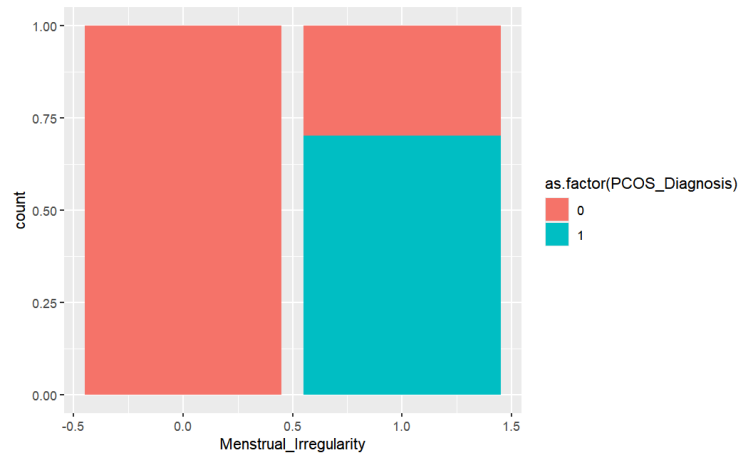
Now I will look at box plots of each explanatory variable against my response variable, PCOS diagnosis.



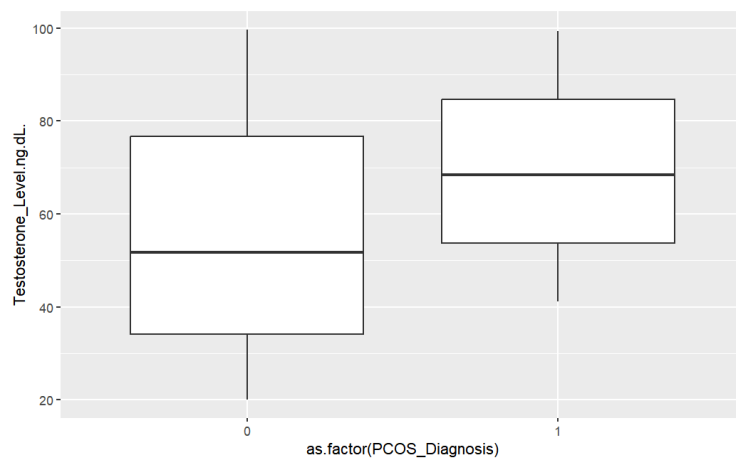
Here we can see that there is a slight difference in age and PCOS diagnosis. It seems women around 30 are more likely to get a PCOS diagnosis while women over 30 do not. But, still, this is only a very small difference and I may find out that age is not significant to PCOS diagnosis.



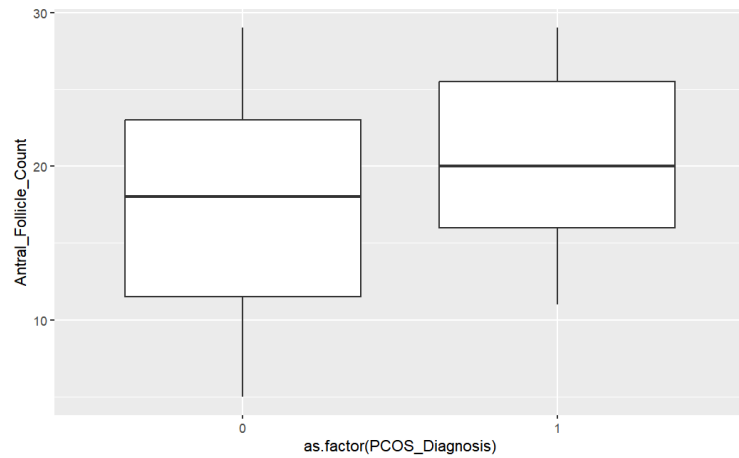
There is an obvious difference in BMI scores and PCOS diagnosis. It seems that the higher BMI score, the more likely you are to be diagnosed with PCOS.



Since menstrual irregularity is a categorical variable, I thought it best to show its effect against PCOS with a bar graph. Here we can see women with no menstrual irregularity (0) do not get diagnosed with PCOS. On the other hand, about 70% of the women who did report menstrual irregularity (1) were diagnosed with PCOS.



Same as BMI, those with higher testosterone levels were more likely to receive a PCOS diagnosis.



Lastly, there is a slight difference in the antral follicle count and PCOS diagnosis. Those with higher follicle count were more likely to receive a PCOS diagnosis. Although, perhaps I am to discover that this variable is not significant.